Supplementary material

Large-scale cortical dynamics across anesthesia and sleep in macaque ECoG

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Supplementary Methods

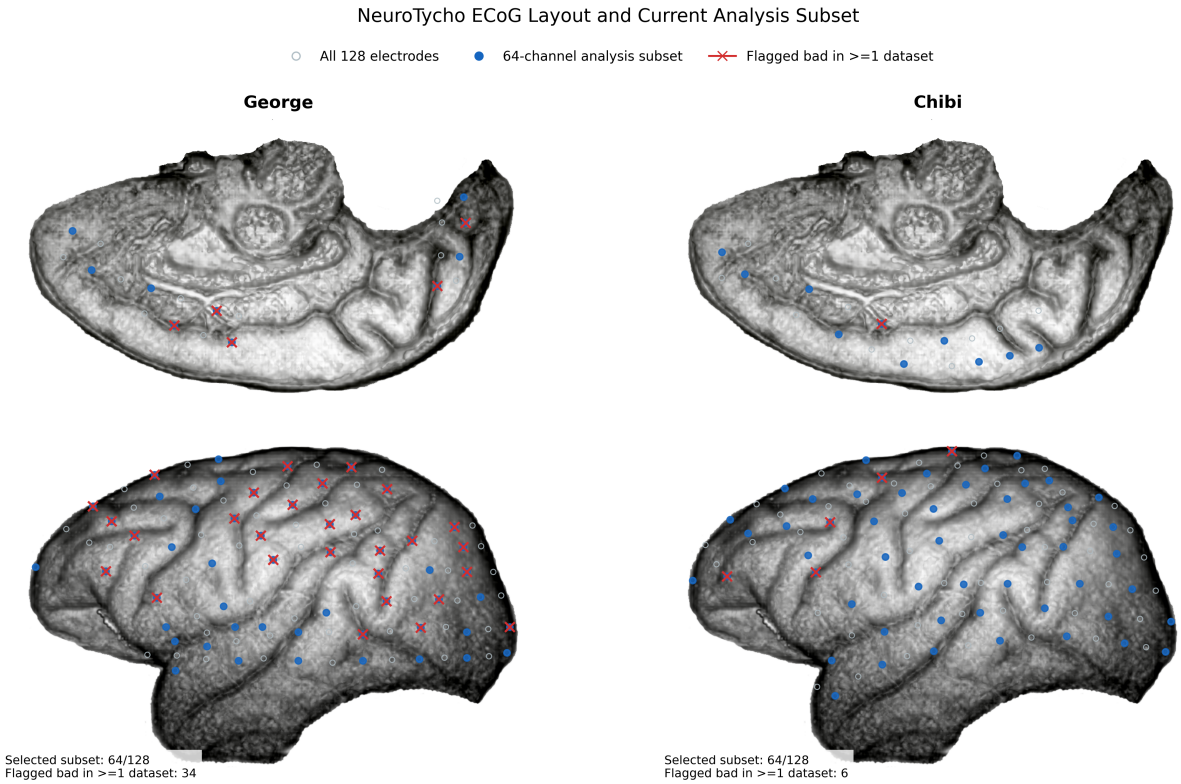


Figure S1: Electrode layout and 64-channel subset. Layout of the implanted ECoG arrays for George and Chibi. Open circles indicate all 128 electrodes, blue filled circles indicate the fixed 64-channel subset used in the main analysis, and red crosses indicate electrodes flagged as bad in at least one dataset.

Table S1: Session provenance of retained epochs. For each monkey-condition pair, one awake epoch and one unconscious epoch were retained from the behavioral labels provided with the source dataset. Start and end times are reported in session-local seconds.

Dataset	Monkey	Condition	Retained awake epoch	Duration (s)	Retained unconscious epoch	Duration (s)
George_KT	George	Ketamine	Session 1, *AwakeEyesOpened*, 0.77–810.58 s	809.8	Session 2, *Anesthetized*, 365.46–964.20 s	598.7
George_MD	George	Medetomidine	Session 1, *AwakeEyesOpened*, 0.90–752.65 s	751.8	Session 2, *Anesthetized*, 692.69–1293.29 s	600.6
George_PF	George	Propofol	Session 1, *AwakeEyesOpened*, 1.13–942.81 s	941.7	Session 2, *Anesthetized*, 615.58–1194.16 s	578.6
George_SLP	George	Sleep	Session 3, *AwakeEyesOpened*, 0.83–658.49 s	657.7	Session 1, *Sleeping*, 37.94–2670.57 s	2632.6
Chibi_KT	Chibi	Ketamine	Session 1, *AwakeEyesOpened*, 0.93–682.95 s	682.0	Session 2, *Anesthetized*, 676.11–1250.12 s	574.0
Chibi_MD	Chibi	Medetomidine	Session 1, *AwakeEyesOpened*, 0.43–626.33 s	625.9	Session 2, *Anesthetized*, 1218.18–2121.62 s	903.4
Chibi_PF	Chibi	Propofol	Session 1, *AwakeEyesOpened*, 32.43–962.23 s	929.8	Session 2, *Anesthetized*, 733.60–1384.25 s	650.6
Chibi_SLP	Chibi	Sleep	Session 3, *AwakeEyesOpened*, 1020.39–1605.14 s	584.8	Session 1, *Sleeping*, 75.44–3000.50 s	2925.1

Notes. Each SessionX folder was treated as a separate recording block with its own local time axis. The analysis retained the longest continuous segment with the target state label for each awake and unconscious epoch. The George sleep metadata file reports four sessions, but only Session1–Session3 were present on disk; the analysis used the available session folders.

Supplementary Results

Table S2: Full statistical inventory for the 64-channel main analysis. d denotes paired Cohen's d computed from unconscious-minus-awake differences. CI denotes bootstrap 95% confidence interval. p_{raw} denotes the exact paired sign-flip p value, p_{FDR} denotes the Benjamini-Hochberg FDR-adjusted p value across all 17 metric-band tests, and BF_{10} denotes the Bayes factor in favor of a state difference.

Metric	Band	d	95% CI	p_{raw}	p_{FDR}	BF_{10}	Evidence
Kuramoto synchrony	delta	-0.827	[-2.92, -0.14]	0.0547	0.1494	2.777	Anecdotal H1
Kuramoto metastability	delta	-0.962	[-2.09, -0.44]	0.0312	0.1328	4.156	Moderate H1
Kuramoto synchrony	theta	-1.092	[-5.24, -0.28]	0.0312	0.1328	6.107	Moderate H1
Kuramoto metastability	theta	-0.792	[-2.80, -0.07]	0.0703	0.1494	2.499	Anecdotal H1
Kuramoto synchrony	alpha	-0.649	[-3.06, +0.04]	0.1172	0.1992	1.655	Anecdotal H1
Kuramoto metastability	alpha	-0.167	[-1.93, +0.46]	0.6641	0.8411	0.608	Anecdotal H0
Kuramoto synchrony	beta	-0.662	[-1.72, -0.03]	0.1016	0.1918	1.715	Anecdotal H1
Kuramoto metastability	beta	-0.167	[-1.38, +0.48]	0.6875	0.8411	0.608	Anecdotal H0
Kuramoto synchrony	broadband	-0.758	[-2.46, -0.06]	0.0703	0.1494	2.265	Anecdotal H1
Kuramoto metastability	broadband	-0.852	[-2.25, -0.19]	0.0625	0.1494	2.989	Anecdotal H1
wPLI	delta	+1.160	[+0.58, +2.63]	0.0156	0.1328	7.432	Moderate H1
wPLI	theta	+0.091	[-0.69, +1.20]	0.8047	0.8516	0.573	Anecdotal H0
wPLI	alpha	-0.116	[-0.81, +0.98]	0.7422	0.8411	0.582	Anecdotal H0
wPLI	beta	-0.260	[-1.01, +0.69]	0.4844	0.7486	0.683	Anecdotal H0
wPLI	broadband	+1.547	[+0.78, +3.87]	0.0156	0.1328	21.284	Strong H1
FCD variance	delta	-0.066	[-0.68, +0.94]	0.8516	0.8516	0.566	Anecdotal H0
Coalition entropy	delta	+0.125	[-1.09, +0.78]	0.7344	0.8411	0.585	Anecdotal H0

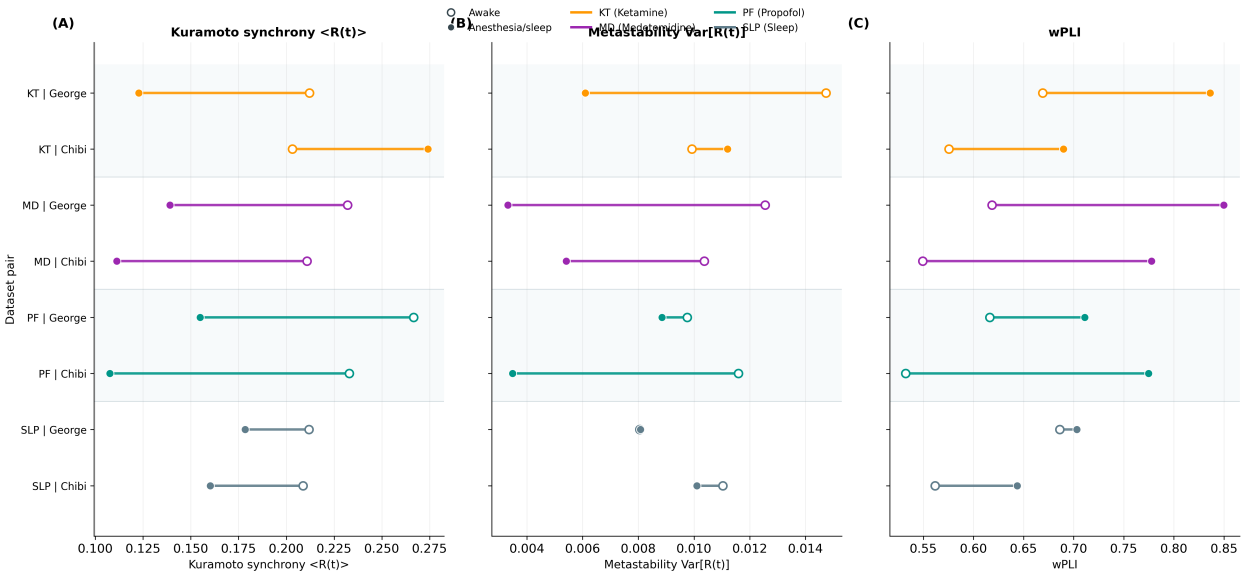


Figure S2: Condition-specific paired shifts. Individual paired awake-to-unconscious shifts for theta-band Kuramoto synchrony, delta-band Kuramoto metastability, and broadband wPLI. Each row corresponds to one monkey-condition comparison. Open markers indicate awake epochs and filled markers indicate the corresponding unconscious epochs.

Table S3: Grouping-aware summaries for the primary effects. Pairwise signs summarize the direction of unconscious-minus-awake differences across the eight monkey-condition comparisons. Condition-mean statistics were computed after averaging the two monkeys within each condition and are included as descriptive summaries.

Metric	Band	Pairwise signs	Condition-mean d	Condition-mean p	Monkey mean differences	Leave-one-condition d
Kuramoto metastability	delta	1+/7-	-1.587	0.1250	George: -0.00479; Chibi: -0.00330	KT: -1.08; MD: -0.71; PF: -0.88; SLP: -1.16
Kuramoto synchrony	theta	1+/7-	-1.395	0.1250	George: -0.0851; Chibi: -0.0514	KT: -2.73; MD: -0.83; PF: -0.80; SLP: -1.03
wPLI	delta	7+/1-	+1.581	0.1250	George: +0.0333; Chibi: +0.0817	KT: +1.19; MD: +0.88; PF: +1.10; SLP: +1.49
wPLI	broadband	7+/1-	+1.783	0.1250	George: +0.115; Chibi: +0.172	KT: +1.33; MD: +1.27; PF: +1.38; SLP: +2.80

Table S4: Duration-matched robustness for primary effects in the 64-channel analysis. The main analysis used retained full epochs. The duration-matched analysis repeated the same metric computation using the first 120 s of each retained awake and unconscious epoch.

Metric	Band	Full-epoch d	Full-epoch p	120-s d	120-s p
Kuramoto metastability	delta	-0.962	0.0312	-0.627	0.1875
Kuramoto synchrony	theta	-1.092	0.0312	-1.247	0.0234
wPLI	broadband	+1.547	0.0156	+1.605	0.0156

Table S5: 128-channel robustness analysis. The analysis was repeated using all 128 ECoG channels before bad-channel exclusion. Rows labeled "Full epoch" use retained full epochs, and rows labeled "120 s" use the first 120 s of each retained awake and unconscious epoch.

Analysis	Metric	Band	d	95% CI	p_{raw}	p_{FDR}	BF_{10}	Evidence
Full epoch	Kuramoto synchrony	delta	-0.777	[-2.95, -0.10]	0.0781	0.1992	2.390	Anecdotal H1
Full epoch	Kuramoto metastability	delta	-0.790	[-1.99, -0.20]	0.0938	0.1992	2.490	Anecdotal H1
Full epoch	Kuramoto synchrony	theta	-1.072	[-5.08, -0.28]	0.0312	0.1771	5.753	Moderate H1
Full epoch	Kuramoto metastability	theta	-0.713	[-2.54, -0.01]	0.0938	0.1992	1.988	Anecdotal H1
Full epoch	Kuramoto synchrony	alpha	-0.620	[-2.52, +0.06]	0.1328	0.2509	1.527	Anecdotal H1
Full epoch	Kuramoto metastability	alpha	-0.105	[-1.67, +0.51]	0.8125	0.9208	0.577	Anecdotal H0
Full epoch	Kuramoto synchrony	beta	-0.576	[-1.60, +0.08]	0.1484	0.2523	1.357	Anecdotal H1
Full epoch	Kuramoto metastability	beta	-0.098	[-1.35, +0.52]	0.8047	0.9208	0.575	Anecdotal H0
Full epoch	Kuramoto synchrony	broadband	-0.717	[-2.49, -0.02]	0.0781	0.1992	2.008	Anecdotal H1
Full epoch	Kuramoto metastability	broadband	-0.702	[-2.05, -0.04]	0.0781	0.1992	1.922	Anecdotal H1
Full epoch	wPLI	delta	+1.087	[+0.65, +2.11]	0.0156	0.1328	6.020	Moderate H1
Full epoch	wPLI	theta	+0.215	[-0.52, +1.29]	0.5547	0.8079	0.642	Anecdotal H0
Full epoch	wPLI	alpha	-0.163	[-0.83, +0.96]	0.6250	0.8173	0.605	Anecdotal H0

Analysis	Metric	Band	d	95% CI	p_{raw}	p_{FDR}	BF_{10}	Evidence
Full epoch	wPLI	beta	-0.191	[-0.87, +0.90]	0.5703	0.8079	0.624	Anecdotal H0
Full epoch	wPLI	broadband	+1.779	[+1.30, +3.45]	0.0078	0.1328	37.138	Strong H1
Full epoch	FCD variance	delta	-0.030	[-0.65, +1.06]	0.9531	0.9844	0.560	Anecdotal H0
Full epoch	Coalition entropy	delta	+0.015	[-0.77, +0.87]	0.9844	0.9844	0.558	Anecdotal H0
120 s	Kuramoto synchrony	delta	-0.648	[-1.88, +0.01]	0.1172	0.2523	1.649	Anecdotal H1
120 s	Kuramoto metastability	delta	-0.626	[-1.45, -0.01]	0.1484	0.2523	1.553	Anecdotal H1
120 s	Kuramoto synchrony	theta	-1.166	[-3.66, -0.44]	0.0312	0.1859	7.580	Moderate H1
120 s	Kuramoto metastability	theta	-0.893	[-2.12, -0.29]	0.0391	0.1859	3.382	Moderate H1
120 s	Kuramoto synchrony	alpha	-0.839	[-3.21, -0.11]	0.0547	0.1859	2.880	Anecdotal H1
120 s	Kuramoto metastability	alpha	-0.480	[-3.33, +0.19]	0.2188	0.3099	1.062	Anecdotal H1
120 s	Kuramoto synchrony	beta	-0.667	[-3.11, -0.02]	0.0859	0.2435	1.739	Anecdotal H1
120 s	Kuramoto metastability	beta	-0.130	[-1.62, +0.48]	0.7344	0.7803	0.588	Anecdotal H0
120 s	Kuramoto synchrony	broadband	-0.656	[-2.10, +0.02]	0.1250	0.2523	1.687	Anecdotal H1
120 s	Kuramoto metastability	broadband	-0.577	[-1.68, +0.09]	0.1484	0.2523	1.361	Anecdotal H1
120 s	wPLI	delta	+0.830	[+0.17, +2.50]	0.0547	0.1859	2.803	Anecdotal H1
120 s	wPLI	theta	+0.344	[-0.44, +1.20]	0.3672	0.4802	0.790	Anecdotal H0
120 s	wPLI	alpha	-0.033	[-0.77, +0.78]	0.8984	0.8984	0.560	Anecdotal H0
120 s	wPLI	beta	-0.522	[-1.52, +0.16]	0.1953	0.3018	1.178	Anecdotal H1
120 s	wPLI	broadband	+1.656	[+1.09, +3.38]	0.0078	0.1328	27.845	Strong H1
120 s	FCD variance	delta	+0.258	[-0.73, +0.99]	0.5000	0.5667	0.682	Anecdotal H0
120 s	Coalition entropy	delta	+0.504	[+0.00, +1.05]	0.5000	0.5667	1.126	Anecdotal H1